

Vicente Raja & Jeff Beck ~ Active Inference Insights 007 ~ The Markov Blanket Trick Debate

Insights | Active Inference Insights 007 ~ The Markov Blanket Trick Debate | 1:56:24

hello everybody and welcome back to the seventh episode of active inference insights I your host darus PV Wayne and today I have for the first time two guests to bring you Vicente Raja is a research fellow at the University of Meria and affiliate faculty of The rotman Institute of philosophy at Western University in Canada his research lies at the intersection of philosophy cognitive science neuroscience and the history of the Sciences of the Mind in 2021 he was the first author of the paper the mark of blanket trick on the scope of the free energy principle in Act of inference which argues that quote the Fe is not the general principle it is claimed to be and the act of inference is not the all-encompassing process Theory it is purported to be either we also have Jeff Beck Jeff is a computational neuroscientist studying probabilistic reasoning in humans and animals he is also the senior machine learning research leader versus and is a critical contributor to the active inference field writing among others the the paper Mark of blanket and Mark of blankets and basian territories guys thank you so much uh for joining me it's an absolute pleasure um you know we're talking off camera there may well be some points of Divergence but we are four days away from Christmas at the time of recording

and I think the baby Jesus would want us
to find some
conciliation so we will do we will do
the most we can to strike at differences
and elucidate differences and see uh but
all in the spirit of hopefully
collaborative science and uh
collaborative thinking so in that with
that in mind I think where we could
start is actually where we have no
issues with one another so looking at
that paper where are the kind of points
of just neutral classd this is like a
good starting
point vient I don't know whether you
want to start given you wrote the paper
but where do you
see where Jeff's work or the other key
contributors active influence where do
you see that you know some of the stuff
they say is actually pretty
uncontroversial good question um so
first of all thank you for for for
having me here um um I think I could
start with a very brief detour on why I
wrote this paper or why we wrote uh this
paper on or
even why the paper is the way it is uh
first of all I wrote I started writing
it to actually to make sense of the FP
and active inference for myself right to
to try to see how it was Drive the steps
and actually section two of the paper
remains there just as that right as
starting with a random dynamical system
and just trying to go step by step there
so this is I think one of the points in
which we converge is at least from my

point of view as a person who works in the philosophy of the cognitive science and the cognitive science themselves that dynamical systems is a good way to look at cognitive systems so that's I think we are in kind of the same camp then uh maybe the modeling strategy that people like me that comes from radical embodied science and ecological psychology to model dynamical systems and the Bayesian twist of it that you find in active inference are slightly different right but you know the this first step this first point I think is a is a point of conversion to um but then after after we we made sense we think of of of of the of the free energy of the free energy principle and active inference ER we our point was okay let's accept all these formalism is good right we don't challenge it we just accept like this is right but even accepting that uh we have some questions about it that how that those questions have to do more on how different bits of the formalism relate to different aspects of cognitive systems and their relationship with the environment right um so and I think this is where the Divergence is stronger so I think the convergence has to do with the with the Primacy I don't know if the primacy if it is the word but you know the attributing importance to a dynamical systems take on all these issues and the the good spirit of trying to find formal methods to study it and our

Divergence I think has more to do with
with uh with with
some conceptions on how to B how to best
address dynamical systems or cognitive
systems in dynamical terms and all that
stuff yeah excellent okay that's that
that's great that's a great start um
Jeff let's go to you let's imagine that
you're reading the mark of blanket trick
for the first time where we will come to
the divergences I promise you well I I I
can tell a story or I can well so I I'll
start in the spirit of this I'll start
off with some with with the point of
agreement that I had with with with the
paper which was one of one of the there
there there are two sort of often
discussed problems with like the FP
literature one is is that it's very it's
it's very it uses idiosyncratic notation
and it can be very difficult to
understand um the other one is I is the
thing that apparently causes a lot of ey
particularly from philosophers and and I
agree that this is a bit of a problem
which is there's a a a penchant for
rhetorical
excess if I could say such a thing which
is a thing I don't think that there's
many many problems with the FP as a
whole but I think that it could be
presented in a more com Compact and
concise way that avoids you know some of
the more philosophical claims that are
often sprinkled throughout these
manuscripts um and I do think that would
create both you know and that this is
something that would create both a bit

more clarity and it's something that
like you know people should poke back to
like when you say like if you find
yourself and this is my personal view
this is not like everyone's everyone who
does this if you find yourself putting
the word thing in scare quotes you might
want to think maybe you're doing
something wrong right it's just sort of
like my knee-jerk
response um and so you know that's that
I think is certainly a point where
there's definitely going to be some
agreement which is is that you know for
me mathematician simplify things be
explicit that's the beauty of it don't
add things to the equations that aren't
necessarily there um but of course
everyone every scientist does this right
we all want to sell our ideas we want to
make we want to make them as big as
possible and so there is some of that
and I think that in particular because
the math is so difficult to understand
and follow right ha you know has um that
those sorts of excesses have become the
Focus right that's easier to sort of
like say oh that's you know this is a
problem this is a problem and so yeah
that's that would be the point of
agreement if I were to quickly and
compactly summarize the point of
disagreement um it's I I will do so by
telling the following story so when I
first came across um Carl FR I was at
the Gatsby at the time and um so Carl
friston was like literally in the
building next door and that's when I

when I first came across him when I saw
him present his stuff and I read a
couple of his papers and and um for me
there were many things that I was like
concerned about let's just leave it at
that let's say concerned we can get into
it more later but the one thing that
wasn't controversial at all in my
opinion was the markof blanket part of
the story and that I think and I and my
hope is is that you know is that as this
conversation proceeds we'll be able to
like I'll be able to explain what I mean
by that um and
maybe come to some sort of consensus so
so I'm happy to do that now but I I feel
like yeah yeah we we we unpick stuff
gradually that's quite that's that's
quite a good irony uh that the markco
blanket was kind of the point of ah this
one this bit works yeah I mean I I take
your point that the maths ought to be
elegant
parsimonious um as much as possible but
unfortunately or or fortunately any gr
unifying Theory comes with it the
philosophy the cognitive science the
Neuroscience the theoretical biology and
so on um and so I think it's an
inevitable consequence that we get
wrapped into epistemological and
ontological questions but we will do our
hardest to unpick them with Simplicity
because I think that's actually all that
we're aiming for so I read the I read
the paper obviously I read it multiple
times and I've read the the commentaries
and and your response to the

commentaries and for me there were about two or three distinct critiques of the free energy principle and active inference that I think we could have a good conversation about the first is that Mark of blankets failed to capture all possible types of conditional independencies so for example candles candle relationship to the external environment or groups of people cultures societies their relationship to uh the whatever those external Dynamics are that they're shielded from by the markco blanket and then linked to that and then I'll leave it there is that these things to use Jeff's quot marks have properties certain things have properties that cannot be captured by the mark of blanket formalism and viente you mentioned relational properties and constitutive self-organization so I don't know whether you want to take those as two separate things one being um the incapacity of the markco blanket to capture all types of conditional independencies um and then those two strands the constitutive self-organization and the relational properties I don't know whether you want to take them separately or whether you think they fall under the same blanket if you pardon the pun okay we'll do it separately Jeff is just pointed to me let's start therefore with um incapacity as I said for the mark of blanket to to capture all conditional independencies and let's use for sake of

Simplicity this uh object that I
selected before the candle what what
does the candle tell us about the free
energy principle viene please um
actually we we use candle Flames because
it is an example that was in the in in
the in the literat to right it seems if
I remember well I haven't I haven't
reread the
the paper so I might be not fully
accurate about what we claim there but
if I remember well the issue with the
mark of blanket is is is that what we
can take as blanket states are are
changing to fasting time so so so
there's no way to to find them and to
and to know and to and to measure them
over time in in in the way that is
required by the
formalism and if I remember well Thomas
par H talks about it in his in his uh
commentary to the to the to the marcoo
Blan treats it's actually the only
commentary that concedes that we might
be writing something in so which is an
interesting thing but it's okay H and he
claims that it could be solved in if if
in in this kind of of systems or in this
kind of things quotes um
um uh we look to instead of looking to
States themselves we can look to to to
densities of of States you know we can
try to model those States and I think
this is an this is an interesting move
that points out why we think that there
are there is more about the marov
blanket uh than just to say oh no it's
just the simp answer to how to

differentiate this you know the internal
and the external States right the the
the the free energy
principle uh and and active inference
doesn't really tell you whether you have
to choose States or densities of States
as the blanket right so you are putting
assumption extra assumptions there on
how you choose to do that and this is
kind of the overall thread of the
critique when we claim the free energy
principle is not the principle it it
claims to be we we are not claiming that
it cannot be used as the as the as the
principle of Le action to you know to to
write down some equations but that it
seems like just from this principle and
from answering two simple question what
is to be a thing and how to to
differentiate from the environment you
have everything and we try to to to
posit examples in which you always need
an extra step you always need extra
assumptions that that are not always
clear in the in when you only appeal to
the formalism so this is the role of our
of how of why we think that the fact
that candle Flames as a as an example
that was in the in the literature why
candle flames are important I think they
point out that you always need you
always need extra assumptions that are
not always clear in the in when you only
appeal to the formalist great um perfect
that's a really nice exposition of of
the claim that you're making uh Jeff
I'll jump to you then um it would be you
know uh R talk to that would be

wonderful and I think actually with specific mention if possible of this dichotomy that viene mentioned which is are we talking about States some kind of ological thing again or we talking about probability densities and where does that line become blurry what actually is that difference or is it just are we just making a kind of false identity claim what what's the sort of what's really going on

here um so I think that the best way the best way to respond is would be so the the the short the short piy and dismissive answer is to say I I solved the pro I solved the flame problem I wish I could share my screen I have some really cool like I have some really cool graphics um so what one of the one of the other problems that that that and when conveying the fre free energy principle that um its its proponents have faced is the fact and I sort of say this the math is really hard and in order to in order to make it tractable a lot of additional assumptions get put in along the way just to make the math easier right and some of those assumptions

are necessary to make you know many of those assumptions are not actually part of the free energy principle by itself right or part of the marov blanket story on its own right but are done to make the math easier like when I first came across the FP I thought there was no difference between active inference and KO control

theory right and the reason why was because when they go to actually solve the equations they make some additional assumptions to make the math easier so that it ends up being equivalent to KO control theory but if you go in and sort of say well what actually is the principle itself as derived either as Carl does it from um uh statistical physics or the way I prefer to do it which is directly from information Theory right that's not the case right the actual like objective function of active inference and all solving these like complicated path integrals and things like that but to make the problem easier to make the math easier to make it more solvable additional like factorial assumptions are built in and a lot of people jump on the fact that like oh it's like you have all these additional assumptions they kind of like but those are done often for like computational reasons they're not necessarily part of the we like to joke that you know you know everybody talks about the generality of the path integral formulation of and and Jan's maxent and how it's mappable on this physics what people often forget is the problem with the path integral formulation is you actually have to solve path integrals which is a serious pain in the butt and so this is sort of like my summary I what one of the things that has contributed to a lot of confusion has been these additional mathematical simplifications that are

then taken to be part of the theory when they are not um in regards to the flame story because I know I gave like a a py and dismissive answer and I'd like I don't you know that was just I will fix that now in the interest of politeness um one the reason that I didn't think that the markov blanket story was like at all controversial was because of its link to systems identification Theory right the Markov blanket itself basically just specifies the domain in which systems identification Theory can apply right so the idea is so what so how does it so when you I don't have you know so system environment inputs outputs right if you're doing systems identification Theory you're basically saying well what is the system doing how what is its input output this is very colloquially right we can make this more what is it it's input output relationship in order to do systems identification Theory right you need to know what that input output relation that's the goal to figure out what that input output Rel and if you see a particular input output relationship that gives you the label for that kind of system right it's a system of Type X because it has this input output relationship the markov blanket is basically just specifies the domain in which you are going to do a statistical version of systems identification Theory right it's the domain of the inputs and

outputs of so when somebody says I have
a big system and there's lots of stuff
in it and I'm going to segregate out a
little system from it the mere fact that
you can do that to when you say like
this is a subsystem you have already
said that there's a Markov
blanket does that make sense right all
subsystems have blankets like by
definition it's it's it's sort of implic
that's and that's why I never thought
that it was controversial because what
do we do we want say well that's a
subsystem that's an important thing it
gets its own label and so it has a
blanket it's like it's it's it's just
it's the way we do systems
identification Theory so I don't know I
want to hear if I want to see if
Vincente has a response before I move on
to the flame in particular I mean that
that's that's kind of
I that's that's that's the that's what I
read when I when I was reading the
literature more uh it's kind of it's
it's it's it's h noncontroversial for
the people who work on it in that sense
like you you you identify a system uh
there therefore there's there's there's
a blanket and there's an external State
I think and part of the Marco blank
trick is to see different issues we find
with that that that for many systems we
are interested in we don't see it to be
so straight forward like we when when we
model with dynamical
systems for instance with the hkb model
H to the the behavior of two couple

oscillator where the where the
collective variable is relative
pH our system is the couple oscillators
so you can put a blanket around it I
guess but then what's the
environment uh it's it's it's the it's
the blanket put on the face of space of
the relative
face which is not a property of any of
the two things there right so I mean I
kind of understand the abstract sense
but then I don't exactly see how this
apply to kind of systems that we are
interested in right so I I I'll just
quickly respond so so the problem with
the whole Markov blanket part of the
story is that there's just two goddamn
many Markov blankets right you have this
big complicated system
right and even you know and even if
you're going to limit yourself as they
often do in the FP literature to like
static Markov blankets right where one
once a particle or constituent component
or whatever has been assigned to either
environment blanket or object it's
always there right so this is a
situation where there's no such thing as
matter transfer right which is why the
flame Case by the way like doesn't work
right the flame the Flames have Markov
blankets but they're not static right
because as the matter you know as it
burns and moves in and fuel comes in and
and exhaust goes out right the matter is
constantly so if if your definition of a
blanket is that it's tight to matter
then yeah you're right like you can't do

a flame at all but that's so but putting that issue aside the problem with Mark blank is is that they're everywhere you have a glass of water every single fluid element in the glass of water so what's a fluid element right you take a glass of water you draw a little circle I mean obviously it's not it's 3D right but but an arbitrary fluid volume has a blanket right and we and we know this like when we do finite element methods for simulating them we draw the we take our glass of water and we make it into lots of little cubes and all the cubes are connected and the boundaries between the cubes form the blanket there's too many blankets blankets are absolutely everywhere right and they and their boundaries are often completely like where you're going to put your blanket is the hardest problem to solve right and I think that that that the free energy literature has not worried too much about the problem of where you're drawing the blanket because the idea was if you've got a blanket then it's subsystems have blankets right we're interested in subsystems and so now we can just like put that problem aside but it's a really difficult problem to say well what is the principle that tells me that this gets a blanket right why why is a blanket that's around you know a bubble Rising through my glass of water the right blanket as opposed to an arbitrary blanket around a fluid element that has no bubbles in it right and I don't think

that that that that question has not been adequately addressed in the you know you know in my opinion but I believe that that there is a notion that is sort of implicit in the Fe approach that handles that and so when when I when I sat down and tried to solve the flame problem I asked myself well what additional principle and this might make vincenta happy right there is an additional principle he was absolutely right I think the principle was implicit um but the additional principle is like how do you choose your blankets well you choose them for parsimony and simplicity right in other words we use the scientific method right you take all the ways that you you have this complicated interacting system you take all the ways that you could draw the you know boundaries in them and there's like an uncount you know not uncountable but there's a really large number of different ways that you could draw your blankets and the way you pick the one that's important is the one that results in the greatest simplification of the of the resulting Dynamics this is basically like mean you can think of this as being field or course graining if you're a physicist right that's what course graining is is that you segregate your system into macroscopic variables or macroscopic components if it's a mechanical system and the choice that you make as to how you're going to do that is to is to basically make the system as simple as possible if you take

that approach then you have if you take
that approach and apply it to the flame
situation in particular that's exactly
what you discover right you discover
that the blanket of a flame right is
just outside the reaction Zone where all
of the exothermic reactions are going on
okay excellent no I think we're agreeing
too much um I think that's going very
well um I love the the flame problem I
like that yeah gorgeous I I really wish
I could share my screen because I like
queued up some videos and like you know
and and had some figures that I was like
just in case the flame case really is
cool I'm gonna publish that soon I'm
just working on I'm trying to get like a
whole I'm working on a
um I took some fake I took some data
from you guys know what particle lenia
well you know everyone knows what
Conway's Game of Life Is So This is
Conway's Game of Life but for
interacting particles and so they get
these particles to do cool things and so
you can put in a random bunch of
particles and they form a cell membrane
and the cell membrane grows stuff like
fella and so we applied it to that
situation as well and what you find is
that the blankets are changing over time
and as they change the identity of the
object also changes which is exactly
what you want right and the reason for
this is because the blanket itself the
blanket alone is meaningless right it
just specifies the domain in which
you're going to do your system

identification or dimensionality
reduction right what really defines an
object's type is the statistics of the
domain right and so as a result you can
watch these cells evolve and as the
statist as the statistics of the various
boundaries in the system because there's
more than one um change so does the
identity of the associated object right
which is really what want this is really
all about labeling stuff right right and
in order to label stuff by type you need
a blanket and the way that you identify
a type is by the statistics of the
blanket okay this is really useful um
because yeah I think we found agreement
with the idea that the Fe needs to
integrate some historicity some Dynamics
and perhaps even changing identities of
you as you've just outlined there Jeff I
think yeah well I think the the critical
thing is is recogn and this is implicit
in everything that Carl writes right
it's it it just needs to be made more
explicit right he says the blanket
defines the phenotype that is a
misleading statement it's true but it's
misleading right the correct statement
is the statistics of the blanket Define
the phenotype right right it's not just
the blanket itself that the blanket
itself is just a domain yeah right it's
just where yeah so yeah well I think
this is a um this problem I think has
started to be remedied within the active
inference literature um because I don't
know if you guys have noticed this as
well but there's been a move to turn the

internal generative model to a
generative model that actually both
encompasses across the mark of blanket
the external and the internal States so
this is something that Maxwell Ram makes
very clear in his um in this year's
paper um physics of belief to put it
roughly um but I think that there is
there is still a slight point of
difference here and I wonder whether
it's just a semantic one

Jeff you still talk about internal
States or the particle the thing the
system and the
environment whereas visente is talking
about coupled oscillators and the way
that I think about this is we can either
have an intrinsic or an extrinsic
relation between let's say external
Dynamics and internal Dynamics now what
does that mean well an external relation
would be something like a causal
relationship so the A and B retain their
identity despite the fact that they're
causally related an intrinsic relation
is something more like a familiar
relation okay if I am uh my you know my
mother is defined as my mother because
of the relationship that she has with me
so in many in many ways I find that this
distinction that you guys are carving
here might be neatly mapped on to that
distinction benter I'm going to start
with you do you think that a mistake
that the fvp makes might be to try and
carve out an external
relation so we've got a enacting
changing B which changes a when really

the focus should be on the Dynamics of the entire system as a whole in the way that sort of Evan Thompson and um those who followed spoke about these kind of couple Dynamic systems where the identity is in the mutual coupling good question I mean I would like to to just point out I think Jeff has pointed uh has pointed out uh what our issue is um and and I agree with with pretty much everything he he has said in that in that regard right our issue basically was that I mean we have nothing against marov blanket blankets as a as a methodological tool as a mathematical tool but we would like to know more about how you select the blanket because we think that depending on how you select the blanket you are interested in you be doing two different things right and and and that and that it's there are it's other parts other fundamental parts of your theory what makes you to to to select a blanket let's say for the organism uh and then the organism has internal States and the environment is the external States or if you are an ecological psychologist like me you want a blanket for the organiz environment system and then everything is internal State and it's not clear what the what the what the external state would be there right you can also choose a blanket on the organiz environment system and a nested one and another blanket it's it's blankets all the way

down man all the way to protons they got
blankets too so as as a mathematical
tool so to speak there are too many
blankets right yeah but how you choose
them reflects what kind of what kind of
fundamental Theory you have in mind and
when I say fundamental Theory I think
about psychology cognitive science right
so

um if you for instance think that the
organism has this input output mapping
and all the relevant computations
happens inside so all the control of
behavior all all the are all the all the
all the perceptual inference is
depending on the internal States you you
are first you

are likely

a cognitive scientist of the mainstream
right and you want the the blanket in
the organism and the external states are
basically

input right uh while if you are an
ecological

psychologist and you think that to
understand the perception for instance
the first step is to see the structures
in the

environment those structures are relevant
in themselves and contribute to
perception as much as the internal
states of the organism there so then you
probably don't take those structures to
be just input right

so my point being here that I totally
agree that there are too many
blankets and that this selection process
depends on deeper

theories that go maybe beyond the free energy Principle as such and you can then you can use the free energy Principle as a method to to develop those those theories right yeah so yeah so this is the I I mean my take on this is so I I completely agree right there there is a sort of hidden additional principle but I think that one of the reasons this got neglected in the free energy literature is because the whole point was

we're always drawing blankets you can draw it anywhere and this Theory still applies right doesn't matter where you put your blanket right I can still I can still treat the system as if it is doing inference over its environment it does you know exact you know that that that was sort of like I think the reason why the the sort of additional like how do you select your blanket problem was neglected was because they were trying to show the principle always works right um that's sort of my my my defense of of of Carl Friston's world is that like there's a reason they skipped the problem because wasn't the important problem right the important problem was it doesn't matter how you draw your blankets you still get you still can apply the rest of the principle so if you decide that that your system is a cell that's fine if it's a collection of cells that still works if it's it and so on

right so our question was then I I think this question was in the Marco blancet

trick too what if your blanket is the is
the is the organism environment system
what are the external states to that
system that the system INF first so to
speak if yeah if if you're if you're
yeah so if your blanket is like the
whole universe I don't I'm gonna go with
it's Christmas I'm gonna go with baby
Jesus okay very good right
no right I mean you know the so I think
that that's counter to the to to the
reason why we want to Define blanket in
the first place and the reason is
because we want to label
things right and the reason we want to
label things is you know is because well
it it simplifies our lives right if I
can look and say that's a cat that's
another kind of cat right then not you
know I can sort of use you know that you
know use those labels because they make
my my life easier right they make
prediction easier about what's going to
happen and things like that right that's
the reason why we want to have labels
and the mark s well you know this is a
potential place so that and the reason I
make this because I do think that
implicit in the whole approach while not
necessarily as explicitly made in the
literature as it should have been is
that the goal is to label things the
goal is to come up with phenotypes and
since the goal is to come up with
phenotypes you can ask the question well
why would we want to do that and the
answer is parsimony right we want a
simple compact representation of our

world and the way we do it is by taking big complicated things made of lots of little parts and giving them a single label that describes a lot of their General properties and features and that is exactly what the mark out blanket does or is capable of doing and so that also tells you which is why like in my world right that's the additional PR it was implicit I'm I want to make it explicit that the way that we choose which blankets to focus on is like the way we do science it's based on simplifying Dynamics for the purposes of prediction and data compression Jeff I may jump in here um and perhaps speak on vincente's behalf I'm very happy for him to jump over me and and and clarify this but I think a question comes up here which is so we're interested in labeling we're interested in describing coupled systems so how come when in the FP literature we draw blankets they so uh coincidentally happen to fall elegantly on what we intuitively believe to be things and their blankets so for example yes you say okay it's because well we see cats and glasses and humans and planets and like why wouldn't we carve blankets around them but why in a sense does it work to carve blankets around them if we're dealing more in information exchange than how the object appears to our sensorium and given that as a mathematician or a physicist what does that say about the mapping between kind of our perception and the

information transfer that we're being
able to describe via the Marco
I think very hard about that question
um because my my knee-jerk response is
to like is to launch into my like what
is science for discussion which is
probably not going to directly address
uh what it is that you what what it is
that you want me to to address
um
when
when the issue is still if the issue is
still like
um let's see so there are two there are
two things going on here right one is
what do we do with the information that
the Universe gives us how do we
categorize it label it and make it
simple and easy to understand and then
the other issue is is well what is NE
you know what is any subsystem
necessarily doing right and those are
very different issues right when we talk
about the when we ask the question
though why do we call this a subsystem
and and you know and then like I'm
trying to cut it in half and like this
not and and this not a subsystem and
that choice is arbitrary right that's
one of the the lessons that I'm trying
to drive home here is is that like you
can draw a blanket around anything the
question is why would
you right having drawn a blanket all of
the free energy mechan the basian
mechanics all apply right but why why
would I draw a particular blanket and it
really is because it's useful to us

right if we could just like have a computer that simulated absolutely everything right and that didn't and it was like I don't know it cost like five cents to run we would totally do that and we would be happy and we wouldn't care about labels right but we don't have that ability right we are trying to construct these you know things that we can wrap our brains around we're trying to simplify things you know to make them parsable and understandable to us and so that's why we would choose a particular set of blankets now but again the point though is and the reason why I think that there wasn't enough emphasis in the free energy literature is that all of once you have established that there's a blanket which you can always do right the rest of it just follows right all the stuff about you know is it effectively doing is it behaving as if it's performing inference on its environment yes right I think the more interesting question that Vincente brings up in the paper which which which I read yesterday that I thought was a really great question um was had to do with with well which is the environment and which is the object right because even this parsimony principle that I just established right doesn't help with that because I just said simplify the global Dynamics like what what you know what what um macroscopic Theory does the best that happens to have this partitioning um and

that doesn't tell you which is the but
this is but this is where I think you
now need to move back into physics land
right so people like I I really like the
work of Jeremy England um and
uh and everyone loves like shinger
little I is it what is life is that the
title of it right um where there is you
know in in in that work there is like a
a definition of how you there is a
notion of how you tell which is the
agent and which or which is the object
and which is the environment which is
the agent and which is the environment
and it basically has to do with energy
flow right so you could sort of say oh
well you know the organism is the thing
that takes in energy and prod produces
enty right and that once you've done
that if you're talking about a physical
system right you know then now you have
something that that does in fact
uniquely identify you know which is
which which you don't get out of then
this is my point right you don't get
that out of a out of a Simplicity
parsimony type
principle right I'm not sure whether you
wanna yes go that second point or
whether you want to loop back to my
first point about the mapping between
the supposedly arbitrary epist logical
tool based approach of the Marco blanket
and actually maybe how that maps onto
onto logically at least what we perceive
I don't know which one you want to start
with yeah I think that that that the the
last uh thing or did this last in

intervention by by by je I think it he
States and I think we we are going to
agree here he he literally States the
Marco blancet trick right that that as
as as soon as you put the blanket on top
of a system right or whatever you choose
with all this uh pragmatic stuff then
you can do Bas inference right you as
you put the blanket you have now the so
to speak the data structure with latent
variables observables
and internal ones to do that and I think
we agree that or we could agree that
this is the Marco blanket trick while
Jeff thinks this is good right uh it's
it's it's it's good we think it's so so
right but that kind kind kind of
be the point right so I think the Marco
blancet trick is
good if you are already convinced for
instance that uh perception is a kind of
inference and then it gives you very
precise mathematical tools to model
that now if you think like
some crazy people like me right
that perception is not something that
you know something that the internal
states of an
organ the internal State states of an
organism do in order to know the
external States so to speak
but is a property of the organism
environment
system then this data structure doesn't
really map very well with what we want
to do or maybe it does but we don't see
how and and and part and part of of our
paper is to is to try to to express you

know to to state these doubts on okay
how could apply this if I think this s
thing
right and uh and then second part uh
which is what the what the um why the
blanket falls exactly where it needs to
to be I think this is a part of first
this is a part of a conversation I've
had like dozens of time
with Maxwell right he he's he's the
person I've talked more about this so
gve talk about this and actually in in
their commentary to the Marco blanket
trick they talk about this uh paper he
has with dton the weak marov blankets
right that they prove that there are a
lot of them right like like as soon as
you have Dimensions you know
and and our question in our response was
yes I mean but the point is not that you
have a lot of them it's how you choose
the right one so how you choose for
let's say for the visual system how you
choose the blanket in the sensory
receptors so to speak right and and I
think Jeff has been touching upon that
right in in in in in some of some of the
things he he has been saying
today before I move back to Jeff could
you just maybe for the audience expound
upon what you meant by this sort of
Distinction um Mak sense when we take
for granted perception as inference I
think that would be quite an interesting
Avenue just to unpick a little bit
further just so audience knows exactly
what you mean by that yeah so
uh long story short uh the common theory

of perception at least since handhold
but most likely seem the cart or even
beyond that is that ER perception is is
a is a form of inference right so the
our mind brain is in the business of
inferring the real environment so to
speak
from impoverished stimulation so what we
get in in in a our sensory systems is
usually ER ambiguous not not not very
informative or not completely
informative and
generally uh insufficient as to know
what is the exact cost of the of that
input right like the typical examp ex is
is that is that you know a a a small
close object can give you the same input
that than a larger object that is
farther away right they can occupy the
same angle in your in your retina blah
blah so the mainstream theory is that
the brain is in the business
of inferring the real properties of the
environment from this impoverished input
and it that's so because it it can use
uh prior knowledge memory experience you
know to build up and good model of the
real
environment ecological psychologists
think ER along with many other people
that come that come at least from
William James William James explicitly
claims that the kind of um of approach
this inferential approach to perception
is pure mythology I I mean it's kind of
the way they wrote in the 19th century
right
but the point being is that for

ecological psychologists the kind of
prior knowledge you need in order for
the inference to work is very difficult
to
justify um so they think
that perception is more of a relation of
Rich information that comes from the
environment and the and the internal
Dynamics and the relation itself is what
uh perception is but we don't need an
inference because we don't need to go
from impoverished stimulation to the
real environment but the the there's a
way to understand
perceptual information that gives you
all you need without the need of
prior knowledge
in order to get grips so to speak with
the Real Environment so it's kind of a
direct undirect stuff or perception as
an inference or perception as an as
information detection now I have to put
again because I've talked with Maxwell a
lot about this so I just want to put
a uh so sort
of um
here right you can still see how the
internal Dynamics change in order to
couple with these and
reach information in terms of some form
of Bayesian inference because Bayesian
inference is yes you know you can you
can describe the the path of a dynamical
system towards an attractor in terms of
a probability density of the State so
there's there's a form to to maybe
there's a form to use inference that is
the Bayesian inference way in which Zan

capture both of them I'm still in an ongoing discussion with maxw about yeah this is this is the as if part of the free energy principle right the math tells us that everything right behaves as if it is performing basy and inference right it doesn't tell us two additional things like how shitty that inference is it could be real like a rock is per forming Basi and inference it's just not doing a very good job um and uh you know but it but that's but that's that that just follows that just Falls entirely from the mathematics up right is that is that it is an as if description right the description of anything you know it is always an available description of a phenomenon that it is that it functions as if it's performing inference right um whether or not it's actually performing inference or doing like solving the right inference problem or or however you want to put that um is not is not explicitly addressed and I think a really interesting question it you know so for example when you know when one applies the the free energy principle in practice there's always this implicit assumption that whatever the thing is doing is it's actually doing inference correctly right there's an optimal inference engine sort of like hidden inside now they can get away with this right because what goes on on the inside doesn't matter right and what defines the object is the what defines the thing

is the statistics of its boundary and so
if you have like a really boring
boundary statistics like you're a rock
right then you don't need a very
sophisticated like like the inference
engine that you would actually use to
sort of say well what is the you know
what does this rock have to infer about
the external World in order to be a rock
and the answer is very little so you
kind of get a lot of these things for
free but it is an as if description
right it's not like you know it's it's
just and you know but like I this is you
know another way to get to the free
energy principles from a purely
information theoretic approach right and
I prefer that approach because all of
this stuff makes a little bit more sense
right you're not sort of saying like the
rock is doing inference you're saying
you know the rock is behaving as it
becomes more clear that this is sort of
an as if argument right as opposed to
it's there's literally an internal state
in the rock that is Computing you know
the probability that the that the water
in the Stream is about to change no it's
not right okay yeah I I had only because
have to in order to be a rock yeah yeah
yeah I um I was did a quite a good
podcast with Adam saffron a couple of
days ago and I said my favorite two
words in the free energy principle are
looks
like um but as as if is that as if as if
is the one that I like to pick up yeah
as if looks like uh no I think that's I

think that's useful

clarification um unless you're a pan
psychist and then they might be coming
for you um what's wrong with pan
psychism I don't understand why
everyone's down on hey I I actually
don't mind that I'm kind of into the
spookier versions but well they might
they might be Keen to perhaps I can't
speak for Philip Goff and everyone else
but they might be Keen to bring into
question the validity of that as if or
looks like but we need not go there um I
wanted to sort of I do want to this was
one of the critiques that I kind of
identified so I do want to just get some
clarity on it um and I I'll actually
quote Vicente here so this is vicente's
response to the commentary
um the extent to which something counts
as a thing depends on the choice of
Market Mark of blanket partition which
can be adjusted according to the needs
of the model there is an ambiguity here
on the one hand it is claimed that
things are what they are by definition
even if boundaries do not real align
with the intuitive ones for ordinary
things on the other hand when the Fe
presents possible examples of things
like brains or cells the mark of blanket
always falls in the right place from the
intuitive point of view so that probably
puts more elegantly the point I was
trying to make
earlier je does that shed any Clarity on
where I was going and Jeff do you think
you do you have a kind of repul to that

I almost hate to do what I'm about to do can you give me the quote one more time my bra I'll give you I'll give you the that was was a lengthy quote and I'm gonna need to hear it one more time to fully absorb and appreciate it I'll do a sort of amended version a shortened version on the one hand it is claimed that things are what they are by definition even if the boundaries do not really align with the intuitive ones for ordinary things on the other hand when the FP presents possible examples of things like brains or cells the mark of blanket always falls in the right place from the intuitive point of view okay so I do have a response to that and that this goes back to the notion that what defines a thing where you get this sort of notion of phenotype is the statistics of the blanket and so that gives you the ability to sort of say oh this is like you know like right now I'm I'm I'm concerned with brains so I all I have to do is ask what is the statistics you know what is these statistics of the of the brain blanket right and then that's the thing I'm currently going to investigate right and so you know what but I do think that this does illustrate the you know one of the issues which we've been talking about which is that like there is this sort of fundament you know if you were going to like denovo start just putting blankets on everything in the world right there isn't an explicit principle present in

the FP that tells you how to do that
right there's this additional principle
which is like you know that that that I
that that led to the you know the
algorithm that I used to discover the
blankets of flames which is like
parsimony right but it's it's not now
now the reason why that that they don't
worry about that is because it's the
implicit in every single thing in the FP
literatures notice that we are labeling
stuff right I have the freedom to label
what I'm interested in and having done
that right that sort of tells me what
the blanket statistics are going to be
um I'll just add one other thing the
reason I prefer like the the information
theoretic derivation of the Fe is
because is is because um the way that
works is you start out with a
probability distribution over blankets
right it's like this is like is and you
take it's a it's a it's a ma it's a
James Max cow or Max ENT kind of
approach and you sort of say like hey
like I have some laws that tell me how
the universe works and those are some
constraints right and then I also have I
want to find I want to find some subset
of my system that has these statistics
those statistics are then definitional
this leads to you know sort of Levan
equation type things right that you can
then solve which tells you how the world
has to be how the individual particles
in the system have to behave if you're
doing it in particle space um in order
for that thing to exist right and it has

to exist because you imposed it as a
constraint when you did your sort of Max
entropy thing

right um if you have it if you have that
in mind what you're really asking when
you sort of when you take this sort of
Free you know when you we sort of we if
you take this sort of

um can I can I use the phrase
ontological potential or is that going to
make people upset right one of the
things that comes out if you if you take
this sort of free approach you're
basically like look these statistics

Define an object

type right and now what I want to know
is I want to know like well what what's
inside what's outside like what kind of
environment can it live in like what
kind of computer you know you know what what
does it have to be made of right you can
just apply that those statistics as part
of a constraint do Max ENT and get all
of the equations that tell you how what
kind of environment this sort of thing
can live in right but the starting point
was like I'm interested in like this
kind of thing right and the reason why
we call it an ontological potential is
because you sort of said like you've
You' started by saying this is my
definition of something and free energy
is literally a potential
function like when you do it in this Max
entropy way it's literally a potential function
so free energy is an ontological
potential but it does start by saying
this is the thing this is the thing I'm

interested in it starts by specifying
those statistics the statistics of a
given object
type do you want to add anything to that
I mean I'm wondering whether from a
philosophical if you're putting a
philosophical hat on what presuppositions
just that kind of as as Jeff
said like we're starting from like this
is a thing what philosophical
presuppositions does that uh
entail yeah I
mean
uh
I'm I'm kind of happy with that right
you choose your thing and you you you
apply the methods you want to so I'm we
we we we really don't have anything with
when I see with the authors of of of of
the paper of course but myself also uh
we really don't have
any
any uh criticism to
that
our uh criticism and actually it was
that paper was published along with a
few more that were talking about that
there's this this this one by J Brun Cal
on their blanket and Miguel Aguilera and
Chris Bley had another one there right
so so to speak if you are a front with
look this is what I took to be what I
take to be a thing or this is my thing
of of interest for these reasons and
then I'm I'm going to apply the Marco
blanket here if the methods work go you
know like cool uh the thing is that
sometimes in the literature these extra

steps or these parts are as Jeff are
just have said not explicit enough so it
seems that everything follows from you
ask what is to be a thing and then you
have a full theory of

cognitive neuroscience and we are like
we don't think so right there are other
other other deep ideas that word there
right and but if we go back to the to
the to that

quote

um it's also not about the kind of the
the the general possibilities of the
inference and the Marco

blanket formalism but more about how it
is presented in the literature sometimes
right so sometimes it is clear that you
put the blanket so in so to speak in the
in the space not in the real space but
in you know you you select some
states and then you put the blanket
there and this is clear for instance in
one paper I think it's Michael kirchof
Julian kerstein Carl fron and I don't
know if I and if I if I forget someone
else but they actually modeled couple
the pendulums right and the blanket
states there where something I'm not
completely sure but where something like
velocity

and yeah ration or position acceleration
these are not the the intuitive ones
right this is not the the the surface of
of of of the pendulum so to speak but
then when you go to any figure in the in
the in in in the literature you see the
blanket that is in the in the sensory
receptors or or or or or in the membrane

of of of of the cell when you
expect the blanket would will be from an
intuitive point of view so our point is
that the these two claims seems to be
not very well aligned right either you
find the blanket in an in an abstract
state of space or or or or in
an space of states or or or or or in an
N dimensional states of space and then
the blanket can be velocity acceleration
jerk whatever you want right or you go
to the more intuitive uh way in which
you know the the blanket of the cat is
you know I guess the hair or something
like that or you find a way which which
I guess is the is the is the is the
ultimate aim right you find a way to
align these two
maybe but I I think the reason that this
happens though is because because the
principle is uniformly applicable to all
systems and you know by and all sub
systems and all subsystems of subsystems
they kind of like glossed over that
problem it's like doesn't matter what
system you're interested in this Theory
applies right and I think that's one of
the reasons why like the issue that
Vincente brings up was basically glossed
over right it's because it it was it
wasn't you know the the the the two
interesting questions from my
perspective that got glossed over are
how do blankets form in the first you
know how do the blankets that we would
like to call blankets because they're
blankets everywhere but how are the
interesting how do interesting blankets

form right what are the what are the
mechanics that that lead to that process
and then why do we say this is an
interesting blanket then that is just a
a drop of or just
a fluid
volume blanking on the easy stuff now
why why yeah why is that just a fluid
why is that an uninteresting blanket and
this is where the whole like parsimony
thing comes in right interesting
blankets have interesting properties
right and they simplify the overall
Dynamics but that but but if you if you
if you start out your paper by writing
that like oh well I'm gonna have this
other principle and it's gonna say this
is a blanket and this is the preferred
blanket right then you sort of if it it
kind of tarnishes your ability to say
that your theory is uniformly
applicable to all systems and subsystems
and I think that that is ultimately the
cause of this of of this particular
confusion you know which is why like you
know know like when I saw this stuff my
knee-jerk response was I why are they
saying a flame doesn't have a
blanket right so my I I I should mention
my my uh the title of my PhD thesis was
like what was it complex Dynamics and
solid flame micro structure oh actually
burning
stuff and and the way that we model
Flames is explicitly with a blanket the
blankets you know is is is there the
flame it's burning this way down a fuse
like this I think Acme bomb right it's

going like this it's going down a fuse
and there's a blanket right the blanket
is basically just outside the reaction
zone right in the same way that like you
know fluid flowing or air going by like
an airplane wing forms a little boundary
which we call curiously enough the
boundary layer right um and you know and
that's a blanket too we are constantly
using blankets because they simplify
Dynamics and that was exactly how I got
to this you know that's exactly why to
me that there was this implicit
parsimony principle that solved that
particular problem excellent um but but
I but I certainly agree like you know it
was it was a rhetorical choice right you
want to show that it it always applies
so of course I'm just going to randomly
draw blankets as you know as need
be because it always
works I I like that you know in the in
the in
the I mean in the in the spirit of
generality they kind of glows over these
tiny details that there are some systems
that might be tricky right but I think
there's kind of
another reason for that I don't think
it's a it's a it's an opposite one it's
a it's a complimentary one but and and
it's a reason that I think applies um
mostly to Psychology and
Neuroscience um like like Jeff I don't
I'm not a super fan of the of what what
they call the high road to to the to the
fre princip yeah the the random
dynamical system plus I like uh better

the one that comes from predictive processing and builds up right first uh because I think it is it it what happened historically right with the with the 1990s paper on how to encoders to you know and you can see that and then uh because it makes explicit how the free energy principle within Psychology and Neuroscience grows within a theory within an overall theory in which this separation is already taken for granted right you see there are the internal states of of of which are the brain that are making the inference there are some blanket States right which are receptors and there's some environment there so right so right this kind of blanket formalism perfectly fit the overall idea that there we have internal states that have to infer the external one so it was not an issue for kind of say the mainstream psychology Neuroscience to think in that way it was like taking for for for for granted so to speak but then this there's this weird Weir weird people call ecological psychologist that have issues with that thing and and and of course that's from that's the place from from where our paper comes right we see issues in that in some systems in that in that so so so I'm gonna propose like The Happy The Happy Middle Ground here right so so the statement that all things behave as if

they are performing inference is kind of hard to get around right because like it it literally follows from I mean it's it it follows from whether you take statistical physics or information Theory it's sort of a statement about like an but what what it doesn't tell you is the thing that is is what it doesn't you know so now you say well I don't think the brain is doing inference or I don't think perception is inference or something like that it turns out that's not incompatible with with with what the free energy principle states either and the reason for this is because the free energy principle says that all things behave as if they are performing inference it says nothing about the quality of that inference it could be really shitty inference it has to be good enough so that the thing can maintain its boundary statistics the the statistics that Define it as like that type of a thing but it could in fact be really bad compared to what a normative model of inference would be would be performing and I think this step this step has also been been sort of this this little bit has also sort of been been missed right there there in a lot of applications because like especially like what I do nowadays which is like you know AI applications you don't want to find systems that do shitty inference right you want to find systems that do optimal imprints right and so but but but in practice right that that becomes the new

question sort of like well okay so they've shown that it's always doing inference and I'll just accept that but I don't care I don't care because I'm interested in the exact in in the kinds of inference mistakes that it's making right it's kind of like saying like when we're in the lab right so we I've worked with a lot of Behavioral scientists over the years and when they're in the lab they always sit the monkey down and okay let's not do monkeys uh there's a mouse and the mouse is doing some task and we always want to ask questions like well how how well is the mouse doing this sort of like this task now we can solve that problem because we designed the task and we set the mouse to solve it right and so the um we can sort of say oh well he's doing he's getting like 95% correct or let's say he's getting doing 90% correct H but the maximum he could given the task difficulty the best he could possibly do is like you know n 98% or 96% correct like that so that's comparing the animals Behavior to Optimal behavior and we do that a lot in the lab but when it comes to really complicated things we don't H we don't have that we lose that ability right and this is where cognitive psychology comes in right cognitive psychology is like okay there's no way in hell a human is going to solve this really complicated social dynamics problem optimally in fact we don't even know how to write down the solution to this really complicated social dynamics problem

properly now you might be able to say
from this free energy perspective that
the person is behaving as if they are
solving it probabilistically or as if
they are doing inference but it's not
optimal inference and so now what what
becomes the interesting area of
Investigation the inter area of
Investigation is what are the mistakes
that some someone is making how do we
characterize those like how do we just
characterize their behavior right we
don't have to you know if you're a
psychologist you don't have to commit to
this as inference problem right because
the the question that you want to answer
is not is it doing inference but it's
what sort of you know what sort of
mistakes are you making what sort of
assumptions are you making not which is
a statement about the quality of your
inference algorithm which is much more
interesting than whether or not you're
doing

inference I

that was way more long-winded so I
apologize I meant that to be Compact and
concise just didn't it's a it's a really
no it's a really useful Edition um so
backtracking 15 minutes just for the
audience um the paper that viente
mentioned is I believe uh kof and
colleagues 2018 the mark of blankets
Mark of blankets of life I like just
clarifying that in case people want to
backtrack and um I believe that's the
that's the blankets of blankets one
right yeah that's the one with the

pendulums

clarify that okay this is I mean again

we're getting on far too well um but no

we're doing well we're doing well no no

one shouting at each other yet um okay

no this is this is super so so as you

said Jeff um we kind of need to do

approximate Bays sufficiently well that

we don't dissipate into the heat bu or

it looks like we're doing that um and

that's I I I feel like some of the ways

that issue got skirted is that we would

encode a higher order uh if we're

talking about humans a higher order

prior preference that we believe that we

should be or we are minimizing

variational free energy so there's yeah

no what yeah is that like kind of uh go

ahead go ahead I'm just curious about we

we don't have to believe anything about

this right organ a cell doesn't have to

believe in the free energy principle the

point is is that it's a mathematical

inevitability right it's it you know you

know so you know would so I guess okay

so let me let me go back there are two

ways to think about the free energy

principle and one of them has to do with

like like self-organization so what does

self-organization mean sort of like it

sounds as if you're sort of saying that

like oh I have a notion of

self right that that I act so as so that

that I act that you know I take action

so that my notion of self is

coherent I don't like any of that stuff

um I know that a lot of other people do

but I like because because the whole

point is is that you don't even need a notion of self right we it you know in order to behave as if you have a notion of self right the the way it's constructed it's sort of like oh okay it has a boundary the boundary has these statistics what simply has to happen to maintain those statistics those statistics give us a notion of object type which you could call self and you could sort of say well okay for now you know then you could sort of ask the question well inside does it also have this notion of self I don't think it's a meaningful question right it's you know it's yeah it's sort of like well we've we we've said like this no you know this is the object type this is its definition and we're really asking a question how do internal and external States collaborate in order to maintain the boundary and if you take the if you take the the max entty you know uh version of the derivation that becomes explicit what you're really asking is how do internal and external States collaborate to maintain boundary statistic has nothing there's no notion of of like self yes I no I'm completely on board with that I do uh some of the work that I like to focus on is actually about self-modeling self- construction um which completely is opposed to the cartesian ego picture and I think actually a lot of the conceptual problems come from a retention of the cartesian ego that

people just feel very it's very
difficult to abandon that picture no I I
was only adding this
because well what you said is very much
in line with what Carl said when he came
on which is that there is no real
existential imperative it's just if you
can see object X at time here and object
X at the same supposed object at t plus
one they've had to have done something
that looks like uh Asian inference so
it's a it's an exponential imperative
only in the sense that we can make the
we can we can kind of say that things
persist through time but there are these
sort of added bells and whistles right
so I you know I'm pretty sure there are
papers now maybe these are earlier
papers which say something like there is
I've read it today there's an a priori
belief now whether this is a
propositional or subpersonal belief I
think is somewhat trivial that I am and
again I'm saying this to make sense I am
the type of thing that will minimize
variational free energy so given that
but given your uh point that we can do
quite shitty dis uh variational basing
inference perceptual inference for one
and AR yeah relative yeah relative to
Optimal but yeah right AR divergences
can be quite big
um what Vicente do you see a way of um
accounting
for the Persistence of the organism
Beyond just like this slow optimization
problem um which as I said kind of
historically at least invoked this high

order prior that I am the type of thing
that will um do desense down free energy
gradients way well I
mean I guess this is a question on how
do things persist in time or how what we
call things persist in time so one thing
is that I don't have any issue with the
with the
mathematical imperative part of it right
that's part of of I mean I can say
something like given this and
this
constraints the uh dynamical system with
an with an attractor will tend to that
to that attractor and that's a
mathematical imperative to right if you
have an attractor with the constraint
may be the system will tell that and
actually you can redescribe that tending
towards the the the attractor in terms
of basan inference in terms of self
self-information minimization whatever
or maximization and free energy right so
I don't have anything about
um the thing is is is that the way uh
the permanence of things or the
self-organization of things without the
belief of it has been uh studied in time
uh is is the kind of thing that uh
touches on homeostasis allostasis or
even autopoesis right an activist
think that the thing what the thing does
is to is to self not not only to self
organize but to self uh produce itself
right the the the in so to speak the
internal state of the systems are
engaged in maintaining the boundary
right that's what

they what they what they do there and I don't think they they they picture that in terms of having to believe that they basically picture that in terms of catalytic or even physiological ways to maintain that right um and and I guess that's that's that's an account of homeostasis too of allostasis when you include the dynamical relationship with the with the with the environment right so now I guess the question is that to what extent or the question we at least we were interested in is to what extent um this physical imperative of the minimization of the free energy or the or the Performing as if making basan inference is able to capture autop pois for instance right and not only autopolis not only the self-organization part but the self- production part um and we and that's another question we have in the yeah so so I have two I have two answers right uh the first is is that when it comes to like maintenance maintenance is baked in right if you if you say my definition of a thing is that it has these boundary statistics and their stationary statistics it's baked in right you know which is totally fine that's your definition of that object type it is stable right and that you know which is great the the the hard question and the question that you know that has been I think un that has gone largely unaddressed in the F literature is how do these things form in the first place

and this is right like you know when you say like oh it's a um you know the focus has always been on homeostasis which has nothing to do with starting from a glob goop right and getting something that is stable right it doesn't have anything to do now there's self-repairs in there right and there is like finding you know H days and balance with your environment but there's nothing that is how do I get from you know a chemical soup to a cell that is the F has has historically you know been completely agnostic on that topic

right um and that's one of the reason why I don't use words like you know alost stasis I also forget what they mean because that's a big word yeah really big

word um but uh but but but but which but but that said there are in fact people working on that particular problem that's why I like to like like so I've mentioned Jeremy England like twice now this he get he get a third mention that's how cool I think is I also just read his book which was which was a lot of fun um and that's probably why it's on my mind so the approach that he's taking is well okay so that's the goal right the goal is to say well what you know what actually you know why is it that these that these complicated things form apparently out of nothing and the math that he uses is is in fact the mathematics of statistical physics and the thing that he and you know and the mathematical object that seems to

motivate this is constructing free

energy minimizing

flows

right great he even has a love I think

it's chapter seven in his book where he

then goes on to say and by the way you

could think of this you know these free

energy energy minimizing flows as doing

a kind of computation about their

environment but he doesn't actually say

as if I actually read through it twice

just to see if he used the phrase as if

and it turns out he doesn't which is

unfortunate but he makes a very similar

argument right is is that is it you know

this free energy minimization principle

can in fact also lead to the formation

of these

things the math is a lot more

complicated right and certainly static

blankets don't apply in this scenario

like the STA a static Mar blanket

doesn't apply right um because the whole

point is that over time it's actually

forming that blanket is constantly

changing in morphing into a much more

interesting set of statistical that that

so the statistics of the blanket are no

longer stationary so a stationary

blanket doesn't um but the same free

energy principle applies the same free

energy principle that shorting called

negative entropy um uh or that was

sometimes referred to as negative

entropy it's it's and and the reason for

this is because it all fall out

ultimately from like statistical physics

okay fantastic well but yeah but but

you're right that has not really been I
I just wanted to reiterate I did have a
point when I started the tirade I just
want to get back to it um you know it is
true that that hasn't been addressed
like how these blankets actually form in
the first place is the free energy
principle in its current form has been
largely agnostic and that's why we say
well take a look what jerem England's
doing it's basically for energy too okay
I'd like to get to that maybe as our
sort of closing uh remark because don't
think there's going to be too much
antagonism or or Discord there I will
raise one more criticism that was um
mentioned in Ezekiel depalo's
commentaries so this is his criticism of
the FP and and and then we can move on
to the fluffy philosophy or statistical
mechanics whatever it is um so aalo has
two critiques in a way of um this uh of
the Fe formulation the first is exactly
what we've spoken about which is that we
have to make a choice about where to
carve up the Marco blanket and we've
covered that until the cows come home
although I'd be always happy to return
to that but his second point is and I
will quote Marco blankets are too strong
to account for the way the internal
states of an organism are often directly
affected by external states without any
statistical mediation or loss of
Integrity as anyone who's ever been
Tipsy after a few beers or felt dizzy
from heat exhaustion can verify
so now we're kind of we've gone to a

completely different formulation which
is just no
vicarious blanket States okay the we're
having Direct effects um Jeff I'm going
to start with you
because I your face is telling me that
you don't like that so much um how does
that sound to your
ears I I'd have no idea I have no idea
what is meant by
that um it's sort of like I mean
and certainly the analogy that was
provided I do not understand I I suspect
that either I mean I'm probably being
obtuse or maybe I haven't I'm not Tipsy
enough given
analogy um but I don't understand how
anything can have a direct effect I mean
you know I I can think of another
scenario that one might appeal to to
make a similar argument which is like
and I always talk about like protons and
electrons they have electric Fields
right and those fields have you know an
effect right but to my mind that just
means you know your blanket is the
field right
yeah do you have any thoughts on
um I
was I was imagining that Jeff was not
going to like that and and and I do
understand why right I mean it's kind of
in in the very definition of marov
blanket is those sets of states that are
always there kind of you know mediating
between the the internal and the
external I guess that what Ezekiel was
trying to point out

there it's something
like it is possible for those blankets
blankets to be leaky sometimes I think
this is an idea that that you find in
the in the wig Mar cover
blankets um uh paper
too uh oh know that that there's no
there's no
absolute Fusion so to speak between the
internal and the and the and the
external States uh and that's why they
are weak right kind of there's not a
full
encapsulation ah this is just an an
speculation on on on what's a a SLE
point there right that those blanket
maybe can be leaky in some way
sometimes um yeah may maybe this is an
issue that I have difficulty
understanding because I've already moved
on Beyond in my thinking like blankets
don't have to be static yeah right
because because the the best criticism
of of the original set of Mark out
blanket stories was your basic your
definition of a blanket or at least the
way you are using a blanket prohibits
matter transfer between object and
environment so like good luck describing
something that
eats you know
right so fortunately you know you know
the the way around that originally was
to just sort of say oh we could have
like made the blanket operate on fluxes
instead of particles and I never really
liked that you know but but now that
we've gotten this issue where we've

basically said oh no look Dynamic
blankets it all still works right all
the math still works out just fine even
when there's matter transfer it works on
flames and so to my if that was the
ultimate issue then I would say that
issue has been resolved but I would
understand why someone would have drawn
that conclusion from the
literature because the literature
focused on problems that were a lot more
solvable like the dynamic so you know
identifying Dynamic markov blankets is a
really mathematically challenging
problem yeah well like whereas
specifying a static one makes your life
really easy yeah and I you know you know
you should never be overly critical of
people in my view for you know simple
mathemat you know mathematically
mathematical
simplifications right you should always
be generous and say like oh I understand
that's not part of the principle per se
that's just the particular use of it you
know my my favorite example of this is
is like general relativity you know the
the equation of general relativity has
like three terms it's like big G little
G and T what they don't tell you is that
hidden behind those three terms is a ton
of math which is why there's only like
one solution to that problem
right and the same sort of goes for the
free energy principle right if you
actually write down what the objective
function for active inference is it's
really hard to compute yeah and so so

they made a lot of assumptions and to
simplify the math and to make the
computation easier right and so you
should always be I always encourage
people to be generous with devaluating
things in which there's there are
equations and then there's
simplification

you got to if you're evaluating the
theory you got to evaluate the original
equations right and not criticize the
simplifications very christmy um maybe
this will help or uh or not or maybe it
might muddy The Waters Another quote now
from The Mark of blanket trick um
without the mark of blanket formalism
the Dual information geometry would be
reduced to its intrinsic component which
is the only one needed for standard
thermodynamics and I've written next to
that on my notes why don't we just use
this Jeff why don't we just use
this because we want I I really do think
because like at the heart of a lot of
this is like

phenotyping it's sort of like the goal
is actually to segment the universe into
little understand into smaller
understandable
pieces right and since if that's the
goal

that's that's sort of you know I sort of
I feel like I addressed this before when
I said like you know if we could model
the whole universe like no problem like
why would we bother to chop it up in the
little pieces and sort of say oh no this
is a car I'm gonna summarize car

Dynamics I'm gonna worry about the little bits and part bits and pieces inside because I could you know why would you do that if you could just simulate the whole thing and it was no problem right so there's sort of this there's sort of an empirical question here like that's why the reason we divide things up the reason we draw blankets around things is to make our lives is to make our ability to understand the universe more tractable right is to make is to make it easier to get right the reason why you know it's like you could always simulate everything if you could you really can't but like if you could you would but you can't right you you're looking for cognitive shortcuts if you're a cognitive scientist you're looking if you're a scientist you're looking for simplifications of complex systems you're looking for macroscopic variables it's it's so it's sort of like there's an empirical reason why you do that vident have you got anything to out of maybe I should have asked exactly what this claim uh entails what what the reduction to an intrinsic component actually looks like actually I that was uh I wouldn't say a quote but that was verbatim almost from from from the from the Dual geometry papers right so it was more than saying more than focusing in the reduction you could do I think what what the what what

the the free energy principal proponents
are saying they're like look this is
just thermodynamics extended a little
bit because we have this this this
blanket that they allow us to do this
this all thing right so that's what what
thing and actually I I'm going to one
more time agree with Jeff in many things
uh uh

and I think that again that was that was
kind of the spirit of the Marco blanket
trick to try to go ourself through the
through other things that you need for
for the for the whole building to be
built so to speak and of course you need
to you need to to choose the the the
thing you are interested in you need to
make assumptions on okay these are the
variables I'm going to be working with
and with that I mean uh eological
psychologists are pretty much the first
people that during the 80s started using
dynamical systems theory to to
Model Behavior actually there's a
there's a there's a 1987 book by Peter
kogler and

Michael arvi called information natural
law and rhythmical Behavior if you go
there and read how they gather from
statistical physics and and all that
stuff you see many similar points than
than than you see now in the in in in
the free energy principle right but the
point being when I say okay we we model
or we you know that
within within this kind of dynamic
cognitive science there's a very famous
model the hkb model of couple

oscillators in which the the the main variable is relative phase why is it relative phase well it's a it's a available very easy to track and when you track it and you and you model it you have a very cool two attractors for these parameters and you change and and you see how the how the two attractors go to one so you you can you can express the qualitative changes of of the of the cing of the two oscillator in a very simple way with just one variable and you know and but why you pick that one well because you know it's it it's easy and and and it model it well right so so I mean

I obviously agree with Jeff on that the point was yes in the literature these simplification sometimes are not clear and yeah it's good to to I think they and and and I also understand yet they are not clear because the aim was generality right so we want this to apply to everything so when when when the couple oscillators guy come but no but talk about my thing okay to that's right yeah we need to put all these other things us to make it work right so kind of thing yeah and that's one of the reasons why I really got to get this paper out so basically it's it's you know if you another way to think about what you know the mark out blanket story is is that it's it's an integral component not just of systems identification Theory but also like macroscopic physics Discovery right so why did you pick the phase difference it

was a convenient variable it encapsulated all of the Dynamics right what if your problem was I just have a complex system and I don't know what and I need to discover what the phase variables should be right and the marov blanket story is one way to do that right it's sort of like I want to segate my complex system into different subsystems and if you have a mark blanket Discovery algorithm right then it sort of does that for you it does macroscopic physics discovery that it identifies that that was the simplest variable to use that makes me wonder out of the bloom so I don't have an answer to this is a pure question that might not have an answer when when people who work with dynamical systems or at least part of them to model modor control right they they they gather a lot of ideas from bursting and from hen and the idea of synergy right like like the brain is able to to control the arm as a Synergy because it's already coordinated uh as a as a as a whole unit there could you put so could you put a marov a marov blanket on a Synergy so to speak I don't know so I think that that so so to some extent and so and this is a I'm I'm going to just sort of say this is a gross generalization of what the Discover of what the blanket Discovery algorithm does but in a in a very real way right it's doing dimensionality reduction right which is basically ident Synergy identification right so why is it that

like you know I can model I can just
sort of look look at all this
complicated stuff I can do well it turns
out right there's really only like a
very small number of variables I have to
track because you know they're all
correlated right there are restrictions
or they're constrained right my wrist is
the same distance for my elbow like and
so on and so that's another way of
character well that means they're
correlated right and Synergy and
correlations are very are sort of you
know are related and so to some extent
what the algorithm actually does or when
you're doing Mark out blanket Discovery
what you're basically saying is what's
correlated what aspects of the dynamical
system are sufficiently strongly related
that I can reduce them to a single
variable right so for example if a rigid
body right I can describe this whole
thing by two points right right I you
know you know I don't have to model
every single molecule in my arm right I
can reduce it to a single Vari and
that's another way of thinking about
like again macroscopic physics Discovery
um right but I also think that's one of
the reasons why like the FP is as it's
formulated is strongly tied to just
empiricism right it's it's it has a lot
to do with just how we do science right
we're we're doing like dimensionality
reduction and looking for principles
that uniformly apply um and and I think
that the focus of the literature has
been on the uniformly AY

thing and now we need to bring it into
the real world of empirical inference
where it's the dimensionality reduction
thing but they are in fact completely
compa amazing uh just again to Lop
around to what I had said so that people
know where to go the quote that I chose
is footnote 12 of the Marco blanket trck
I told you I read the paper um and I
think it pertains to this two
complimentary ways of describing the
density dynamics of internal States one
being this intrinsic uh change over time
towards uh the attractor State and then
the other one is this
extrinsic um tracking of external States
and I'm not a mathematician but it
appears to me that the maths seems
equivalent um and we can actually make a
kind of formal equivalence between those
things which is quite quite beautiful
actually it's it's almost as if they're
equivalent I'm not a mathematician give
me a break um no no no no no sorry that
was that was not a criticism that was a
terrible joke oh okay f f oh right yes
yes very good very good it looks like
they're the
same um yeah excellent so I will finish
with what I promised we'd finish with
which is a kind of bigger potentially
philosophical question that expands
beyond the scope of the free energy
Principle as is right now um and this is
an issue that actually is really matters
to me uh when I first discovered active
inference I remember tweeting two things
one of which was uh the presence of

Markov blankets upon Markov blankets is the best argument from creation that's ever been and the second one was um I think that the apparent distinctions of internal and external States mediated for a mark of blanket might itself be an illusion and there may just be some sort of singular reality I'm paraphrasing my tweets but it was more just philosophical eal thinking and we mentioned pan psychism earlier um and again I don't want to get too into the Weeds on this but Jeff maybe we could start with you where do you think this all comes from these um these distinctions that we can carve for instrumental reasons which seem to you know which do work mathematically and what are your hunches about what might lie underneath them I love questions like this and I have the perfect answer so what lies underneath nothing oh that that was my serious face that doesn't help no no so so I remember one time I got asked a question like you know so we're talking about like you know p's and q's because that's what you're doing when you're doing probabilistic reasoning and somebody who raises their hand like is there a true P the answer is no no no there's not and it's not because there isn't a true PE right it's just I don't understand the question right it's sort of a it's sort of a jerky thing to say I I often describe when I when when I get so as you might imagine given what I do for a

living I get asked questions like that all the time and I usually refuse to answer and the reason I refuse to answer is because I got up to work today I put on my coat and tie and I put on my work pants right my work pants are my harsh instrumentalist pants and then if you want me to answer like philosophical or religious questions it's like well I have to put on a different pair of pants right you know when you know when functioning in this in this environment like you know it's it it you know I I take a very very instrumentalist view right and I think the you know so you ask like oh do we carve you know is there some is it all just one thing right are these artificial carvings AR the artificial carvings are artificial right we carve up the world because it's convenient right because it simplifies the problems we have to solve it has empiric it has predictive power and empirical utility

I like to say that science is not about truth it is about prediction and data compression and nothing else right so when functioning as a scientist that's what I focus on which is my way of saying like that's that's my answer harsh

instrumentalism harsh instrumental means you don't have to worry about like what's true I can't wait for all the philosophers to be shouting at you virtually um
viente what are your thoughts on this I was expecting this answer from

Jeff because my preparation to this
podcast was not to reread my own paper
but to watch another podcast on the
machine learning Street talk I think and
yeah that he was saying electron is just
the name for an equation you know
like right so so I was expecting heart's
instrumentalism I kind I'm kind of an
I'm I'm I'm kind of an instrumentalist
too in some ways
uh but at the same time I'm a realist so
how can that be how can that be the case
right I I guess it's it it comes overall
um comes all to to a form of pragmatism
in the end in which you can be realist
about the kind of things you
do uh and if they are successful well
you you you are dealing with the with
the with the real world and um you do so
by using
instruments and I I'm in favor of
aurality of
instruments uh and marov blankets are
one of them that I I don't personally
use in my own research right but you can
use it um but I wouldn't try to find a
final Foundation like oh there's only
one
um or or you know or everything is such
or everything is such other thing right
it's more like uh we act around um
and reality is somehow the way all where
all our
actions
converge um and not actually sometimes
do sometimes don't and and we we can
reality in that way and all we have is
our our Concepts that are instruments

and are you know not foundational in any
way tools marov blankets other tools and
we go like that but I mean I don't think
there's that there's a there there
there's some form
of you
know matter of fact or fact of of of of
the matter so uh so speak so there's
something that that there's something
we some reality that we build up
together through action and is there
it's just it can change so it's not the
final thing it's it's it's it's not the
an old philosopher you know an ancient
philosopher would say something like the
the AR right this this this final
original thing there's no but there's a
thing we do there's there's a there's
there's a matter of fact and and theep
variety of tools and their superposition
and their obus OB obus I don't know that
that that word in English but even the
even the things that they don't match
well tell us thing about what what kind
of thing we can do and we can
do oh where are my Buddhist friends
where are I where my well so that's
that's the funny you asked the wrong
question if you wanted me to put my put
on my Quaker Buddhist pants I can do
that too oh I just generally refuse to
maybe on Christmas day I'll ask it's one
of those it's like
I I remember one of my uh my P one of my
uh one of the faculty in my department
when I was getting my
PhD once said that
um uh philosophy is a lot like like

algebra or or or going to the bathroom
it's just not something you should do in
public I don't agree with
that you
know it it indicates like why it is I
hesitate to answer these questions
because I don't really have good answers
I have my answers and Def faulting to
instrumentalism is like just the easy
way out right sure well yeah they're
purposefully quite provocative and
probing uh but people seem to like
answering them chaps that was uh that
was tremendously fun um maybe we could
have just a couple minutes for closing
remarks um given we've reached such a
beautiful point of conciliation viene
um maybe in addition to sort of closing
remarks you could also let people know
what you're working on and where they
could reach out to you if they wanted to
know more about your topics of
Interest okay so I'm going to
start by by talking one one last time
about the marov blanket trick the paper
kind of grew on its own it's way long I
think it's 24,000 words so it's it's way
it's way longer than I was
expecting uh uh because actually most
most of it is just uh me asking a qu not
me we all of us right but asking the
questions we have right oh but if if the
formalism work why this and why that and
why candle flames and why you know that
kind of thing but I really don't have an
strong uh position on those things I I
just would like to know okay how would
you apply this to a model with a

relative face and that that seems to include the whole system I only have a strong opinion with one thing and which is uh the fact that for perception you need or not need prior knowledge right that's that's that's actually the real my my real State there like I think uh every time you endorse the system with some prior knowledge in order to make sense to go as as we talk from the from the impoverished impoverished input to the real world so to speak uh you have no way to justify that that that knowledge or to explain what's why it is there right the typical way you know what say no it's Evolution right but it's kind of unsatisfactory for many I don't want to go to this thing right but this is actually my only kind of my my Ideal breaker I think theories of perception should then appeal to PRI your knowledge to explain it and I think one theory that that does that is ecological Psychology by virtue of the lawful relationship between the patterns of light and the environment you have in those patterns of light information enough as to know exactly what the exactly to to to to to a good the degree what the what the real environment is with without the need of pulling other kind of knowledge into the equation and I work on that right so so so so part of my work is is is trying to make sense of

a cognitive Neuroscience that starts with ecological psychology for instance if the brain is not in the business of inferring the real causes of impoverished stimulation or of enriching impoverished stimulation with other knowledge as to build a model of the environment what the hell is the is is the brain doing right so within the ecological approach the word uses resonance the brain resonates to this richer information and part of my theoretical work is to try to operationalize the resonance metaphors so to speak so how can we have a a concept that we can actually test and part of my empirical work I'm kind of maybe 70% a philosopher 30% a cognitive scientist cognitive neuroscientist is to try to test that so this year I've been in in in in the mobile lab which is at the technical University of Berlin and they and they specialize in Mobile EG and so I've been Gathering some multimodal data that that that has you know eye tracking motion capture and EG to try to find what kind of patterns of information our brain can be resonating to so how the brain Dynamics can be constrained by this specific structures and not just by by kind of say simple input and that kind of thing so I'm trying to yeah last sentence I'm trying to do something like an ecological Neuroscience uh both in the theoretical

realm and the and in the
experimental uh realm very cool very
cool Jeff go
ahead uh
so I'm taking the exact opposite
approach I'm trying to see how far we
can push this whole prior
thing I I like to joke that the the
difference between a basian and a non-
basian is honesty right is you have
priors right I don't know where they
come from I don't know that they're
normative but they're there you you're
kind of stuck with them all knowledge is
like llas all knowledge is based on um
all knowledge is probabilistic right
it's always based on assumptions there
are two sets of assumptions that are
critical to worry about one is I'm going
to get to what I'm doing in a second one
is the data right any conclusions you
draw is conditioned on your observations
but there's another important thing that
we often forget about right is that all
the conclusions draw are also
conditioned on your models for the
universe and I say models this is just
my view I could see vente is going oh my
God no um when I say models I actually
think the most critical thing that we
often neglect when we talk about the
basian brain is that there's not one
model there's a lot of them and they and
and they and and you don't just use one
at a
time right you use lots of in fact you
one you could even think and this is
very coarse I'm I literally am gonna get

to what I'm doing in a second it's very course but you could think of a Bayesian brain as having brain regions right and don't overparse this because I know people will right and each region is a model right each region describes is a statistical model of something and when you get a new task what do you do right well you have to figure out which collection of models and how they're connected right are the ones that you know are the ones that like solve your problem or model or model the data or describe the data or compactly summarize this like the data compression compactly summarize the information that you received or give you the predictions about the things you care about or however you want to phrase it and so what we're trying to do right now um is we're exploring sorry we're exploring the extent to which we can get this to work at you know at scale so we want to construct a brain that's not like some fast feed forward neural network right but one that is more brain like right by summarizing what by saying that no no no brain has lots of there's lots and lots and lots of models and the way that the brain actually attacks problems is by reasoning by analogy this model explains this part of what's going on this model explains this part of what's going on and the trick is how do you figure out how to connect them and so this is a massive model comparison SL Bayesian graph neural networking type problem which is like computationally intensive like

that's

a i' I'd put that in quotes like but it
would like it would be like several sets
that's that's how much scare it is um
that's what we're working on right we're
working on pushing you know seeing how
far we can really push this basum brain
hypothesis but not just sort of saying
there's a prior by saying there's lots
of them and they interact in interest
way and at that point is it still a
prior viene or has it become something
else right I

mean uh this is kind of the of the
agreement I I I I read with with M well
for instance but with other people there
right yeah I mean and I think it is good
I I don't like the the uh the idea of of
of prior knowledge or one prior or many
of them I think there are are other ways
we like to talk about

ecological constraints that are not
inside here out there well but blah blah
blah blah blah

but what I try try try to do is to is no
priors let's see how far I can go and
it's fa that you guys do okay this this
set of of of priors let's see how far
and that's it that's I think that's good
for science actually yeah no that's the
whole point right because eventually
what we want to do is basian model
comparison between your approach and
mine because you know basian is the only
way to do it

obvious I think you guys are gonna
absolute pleasure absolute pleasure
chatting with you well it's no it's been

an absolute pleasure to witness it um
thank you guys for coming on the podcast
and uh I'm sure there are some people
out there devilish people who expected
pots and pounds to be thrown across
webcams but um it I you know this is
what science is about at the end of the
day um finding convergent paths and
trying to get closer to some
approximation of Truth so I hopefully
we've contributed sorry no and you both
Lean Back simultaneously when I said
that word um okay some something that
helps us predict and data comp there you
go Jeff um but thank you guys again from
me and The Institute and uh uh yeah
everyone listening definitely check out
Jeff and vent's work so thank you again
thank you